

May – July 2026

Nuclear #1 & #2 – Missed-Chapter Catch-Up Plan

Restructured for the May 5 – July 11, 2026 study window

OWNER	DURATION	SCHEDULE	DEADLINE
Parker	10 weeks	Mon / Tue / Wed	Jul 11, 2026

Weekly Cadence

MON	TUE	WED	THU – SUN
4 hrs reading	4 hrs reading	2 hrs problems	off / buffer

Daily Reading Workflow (Mon & Tue)

Each 4-hour reading day runs as a single 3-stage cycle over that day’s assigned chapters. Don’t sequence chapter-by-chapter – batch them through the pipeline:

STAGE 1: Skim + note questions	STAGE 2: AI LaTeX notes	STAGE 3: Reread + answer
1–2 hrs. Fast pass through day’s chapters; jot down concepts you don’t grasp as questions.	30 min – 1 hr. Feed your skim notes to an AI; get a structured LaTeX summary to study from.	~1 hr. Second pass targeting your own questions; write answers in the margins.

Total active time: ~2.5–4 hrs. Remaining time in the 4-hr block is buffer. Lighter early weeks let you settle the workflow before the heavier mid-plan reading load.

Reading Sources

Short	Full reference	Chapters
Loveland	Loveland, Morrissey & Seaborg – <i>Modern Nuclear Chemistry</i> (Wiley, 2nd ed.)	8, 9, 10, 11, 12, 14, 18
Mann	Robert Mann – <i>An Introduction to Particle Physics and the Standard Model</i> (CRC, 2010)	3–6, 9–14, 19–25
LB	Lancaster & Blundell – <i>Quantum Field Theory for the Gifted Amateur</i> (Oxford, 2014)	7, 12–13, 16–20, 26, 46–47
Thomson	Mark Thomson – <i>Modern Particle Physics</i> (Cambridge, 2013)	13, 17 (June companion)

10-Week Overview

Original 8-week chapter plan redistributed across 10 weeks. Loveland material spread over Wks 1–4 (lighter ramp-up); Mann/LB/Thomson material on its original cadence (Wks 5–10). Capstone topics get full dedicated weeks at the end.

	Dates	Theme	Reading (Mon+Tue)	Wed Qs
Wk 1	May 4–6	Beta + Gamma decay	Loveland 8, 9	3
Wk 2	May 11–13	Nuclear reactions	Loveland 10	3
Wk 3	May 18–20	Fission & nuclear astrophysics	Loveland 11, 12	3
Wk 4	May 25–27	Reactors, accelerators, detectors	Loveland 14, 18	3
Wk 5	Jun 1–3	Particle-physics foundations	Mann 3, 4, 5, 6, 9, 10	6
Wk 6	Jun 8–10	QFT: propagators & S-matrix	LB 7, 12, 13, 16, 17, 18, 19, 20	6
Wk 7	Jun 15–17	Gauge invariance & QED	Mann 11, 12, 13, 14	4
Wk 8	Jun 22–24	Weak interactions	Mann 19, 20, 21 + LB 26	4
Wk 9	Jun 29–Jul 1	★ Higgs mechanism & EW breaking	Mann 22, 23 + LB 46, 47 + Thomson 17	5
Wk 10	Jul 6–8	★ Neutrino oscillations & capstone	Mann 24, 25 + Thomson 13, 17.8	5

■ Gold = Wk 9 (Higgs). ■ Green = Wk 10 (neutrino oscillations).

Final deliverable: the 1-page capstone connecting gauge invariance → EW unification → Higgs mechanism → neutrino masses, drafted in the Wk 10 buffer days and finalized by **Sat Jul 11, 2026**.

MAY – Nuclear Foundations

Loveland chapters spread across four lighter weeks. Use the early weeks to lock the Mon/Tue/Wed rhythm and tune your AI-LaTeX-notes pipeline before the formalism-heavy June.

WK 1 Beta & Gamma Decay

May 4 – May 10 • 10 hrs

Goal: re-establish intuition for the two electromagnetic/weak decay modes. ~48 pp – the lightest week. Use Mon/Tue buffer time to set up your AI-notes workflow.

MON Mon May 4 • 4h • reading Loveland Ch. 8 Beta (β) Decay 21 pp • ~2h Loveland Ch. 9 Gamma (γ)-Ray Decay – first half 14 pp • ~1.5h	TUE Tue May 5 • 4h • reading Loveland Ch. 9 Gamma (γ)-Ray Decay – second half 13 pp • ~1.5h Setup Lock down AI-LaTeX-notes pipeline; refine skim-and-question template ~1.5h Buffer Catch-up / re-skim Ch. 8 trouble spots ~1h	WED Wed May 6 • 2h • problems Loveland Ch. 8 Problems 3, 5 Loveland Ch. 9 Problem 2
---	---	---

END-OF-WEEK CHECK

Sketch the β -decay continuous spectrum and explain why the antineutrino is needed. State the multipole selection rules for γ transitions and which is suppressed.

WK 2 Nuclear Reactions

May 11 – May 17 • 10 hrs

Goal: a single dedicated chapter – Loveland 10. Read deeply, with full attention on cross-sections and reaction kinematics. ~48 pp + an end-of-week mini-review.

MON

Mon May 11 • 4h • reading

Loveland Ch. 10

Nuclear Reactions – §§10.1–10.4
~24 pp • ~3h

AI LaTeX notes

Distill §§10.1–10.4 into structured summary
~1h

TUE

Tue May 12 • 4h • reading

Loveland Ch. 10

Nuclear Reactions – §§10.5+
~24 pp • ~3h

Reread

Target your own skim-questions;
margin answers
~1h

WED

Wed May 13 • 2h • problems

Loveland Ch. 10

Problems 3, 5

Loveland Ch. 10

Bonus: redo a problem with conceptual notes in the margin

END-OF-WEEK CHECK

Write the differential cross section $d\sigma/d\Omega$ for Rutherford scattering from scratch; compute a reaction Q -value; explain the difference between direct and compound-nucleus reactions.

WK 3 Fission & Nuclear Astrophysics

May 18 – May 24 • 10 hrs

Goal: two complementary chapters – the same nuclear physics applied at two scales (terrestrial fission vs. stellar burning). ~62 pp.

MON

Mon May 18 • 4h • reading

Loveland Ch. 11

Fission
30 pp • ~2.5h

AI LaTeX notes

Fission summary; flag asymmetric mass distribution
~1h

Buffer

~0.5h

TUE

Tue May 19 • 4h • reading

Loveland Ch. 12

Nuclear Astrophysics
32 pp • ~2.5h

AI LaTeX notes

Astrophysics summary; pp-chain, CNO, r/s -processes
~1h

Reread

Target unresolved skim-questions
~0.5h

WED

Wed May 20 • 2h • problems

Loveland Ch. 11

Problem 4

Loveland Ch. 12

Problem 3

Synthesis

1 short prompt: compare a fission reaction to a stellar burning chain in terms of Q -value sign and binding-energy curve position

END-OF-WEEK CHECK

Explain asymmetric fission-fragment mass distributions; sketch the binding-energy-per-nucleon curve and mark where fission, fusion, and the iron peak sit.

WK 4 Reactors, Accelerators & Radiation Detectors

May 25 – May 31 • 10 hrs

Goal: close out Loveland’s applied/experimental chapters. ~84 pp – vocabulary-heavy but largely descriptive. Lock in detector-vs-particle mapping for sPHENIX work later.

MON

Mon May 25 • 4h • reading

Loveland Ch. 14

Reactors & Accelerators

*44 pp • ~3h***AI LaTeX notes**

Reactor types + accelerator types reference card

*~1h***TUE**

Tue May 26 • 4h • reading

Loveland Ch. 18

Radiation Detectors

*40 pp • ~3h***AI LaTeX notes**

Detector → particle/energy mapping reference card

*~1h***WED**

Wed May 27 • 2h • problems

Loveland Ch. 14

Problem 3

Loveland Ch. 18

Problems 2, 4

SUN MAY 31 CHECK – LOVELAND DONE

Pick the right detector for a given particle and energy regime; describe a thermal-reactor fuel cycle in 5 sentences. This closes the Loveland nuclear-chemistry leg – you are ready to enter Mann’s particle physics.

JUNE – Particle Physics & QFT Foundations

Mann’s particle-physics formalism (Wks 5, 7, 8) interleaved with the LB QFT primer (Wk 6). Each week is the original syllabus’s pacing, now starting one week later thanks to the Loveland redistribution.

WK 5 Particle-Physics Foundations (Mann)

Jun 1 – Jun 7 • 10 hrs

Goal: Nuc #2 territory begins. Six short Mann chapters (~96 pp). Formalism-heavy – this is the math foundation everything else rests on.

MON Mon Jun 1 • 4h • reading Mann Ch. 3 Symmetries 17 pp • ~1.5h Mann Ch. 4 Conservation Laws 12 pp • ~1h Mann Ch. 5 Particle Classification 15 pp • ~1.5h	TUE Tue Jun 2 • 4h • reading Mann Ch. 6 Discrete Symmetries 16 pp • ~1.5h Mann Ch. 9 Scattering 16 pp • ~1.25h Mann Ch. 10 A Toy Theory 20 pp • ~1.25h	WED Wed Jun 3 • 2h • problems Mann Ch. 3 Q1 Mann Ch. 4 Q2 Mann Ch. 5 Q2 Mann Ch. 6 Q3 Mann Ch. 9 Q2 Mann Ch. 10 Q3
---	---	--

END-OF-WEEK CHECK
Derive Noether’s theorem in one line; list CPT assumptions; draw a tree diagram in Mann’s toy theory.

WK 6 QFT: Propagators, S-matrix, Scattering (LB)

Jun 8 – Jun 14 • 10 hrs

Goal: eight short LB chapters (~66 pp). Each introduces a new machine. Take it slow on the propagator and S-matrix chapters – those are the load-bearing ideas.

MON

Mon Jun 8 • 4h • reading

LB Ch. 7

Example Lagrangians

*6 pp • ~0.75h***LB Ch. 12**

Canonical Quantization Ex.

*8 pp • ~1h***LB Ch. 13**

Many Components / Massive EM

*9 pp • ~1h***LB Ch. 16**

Propagators & Green's Fns.

*9 pp • ~1.25h***TUE**

Tue Jun 9 • 4h • reading

LB Ch. 17

Propagators & Fields

*10 pp • ~1.25h***LB Ch. 18**

The S-matrix

*10 pp • ~1.25h***LB Ch. 19**

Feynman Diagrams

*12 pp • ~1h***LB Ch. 20**

Scattering Theory

*7 pp • ~0.5h***WED**

Wed Jun 10 • 2h • problems

LB Ch. 17

Exercise (17.1)

LB Ch. 18

Exercise (18.1)

LB Ch. 19

Exercises (19.1), (19.2)

LB Ch. 20

Exercise (20.1)

LB Ch. 7

Exercise (7.1) – bonus

END-OF-WEEK CHECK

From memory, write the Feynman rules for LB's Yukawa theory and derive the tree-level cross-section.

WK 7 Gauge Invariance & QED

Jun 15 – Jun 21 • 10 hrs

Goal: bridge LB's toy work into real QED. Four Mann chapters (~60 pp). Ch. 12 (gauge principle) is **the** chapter of the week – it reappears in Wks 8, 9, 10.

MON

Mon Jun 15 • 4h • reading

Mann Ch. 11

Wave Equations

14 pp • ~1.5h

Mann Ch. 12

Gauge Invariance

9 pp • ~1.5h

Mann Ch. 13

QED (start)

~8 pp • ~1h

TUE

Tue Jun 16 • 4h • reading

Mann Ch. 13

QED (finish)

~11 pp • ~2h

Mann Ch. 14

Testing QED [skim]

19 pp • ~1.5h

WED

Wed Jun 17 • 2h • problems

Mann Ch. 11

Q2

Mann Ch. 12

Q1

Mann Ch. 13

Q3

Mann Ch. 14

Q1

END-OF-WEEK CHECK

Explain gauge invariance in one sentence, write the QED Lagrangian, and calculate a tree-level QED scattering amplitude.

WK 8 Weak Interactions

Jun 22 – Jun 28 • 10 hrs

Goal: build the weak-interaction apparatus for Wk 9's unification. ~71 pp. End Tue with LB 26 to preview spontaneous symmetry breaking.

MON

Mon Jun 22 • 4h • reading

Mann Ch. 19Beta Decay $\rightarrow$ Weak

22 pp • ~2h

Mann Ch. 20

Charged Leptonic Weak

16 pp • ~2h

TUE

Tue Jun 23 • 4h • reading

Mann Ch. 21Charged Weak of Quarks/Lep-
tons

24 pp • ~3h

LB Ch. 26

Broken Symmetry (preview)

9 pp • ~1h

WED

Wed Jun 24 • 2h • problems

Mann Ch. 19

Q3

Mann Ch. 20

Q2

Mann Ch. 21

Q2

LB Ch. 26

Exercise (26.1)

END-OF-WEEK CHECK

Sketch the Feynman diagram for neutron β -decay with a W -propagator; explain what the CKM matrix does; state Goldstone's theorem.

JUNE 29 – JULY 11 – Capstone Topics

The two requested anchor topics, each on its own dedicated week. Wk 10 ends Wed Jul 8; the 1-page capstone synthesis is finalized in the Thu–Sat buffer ahead of the Sat Jul 11 deadline.

WK 9 ★ Higgs Mechanism & Electroweak Symmetry Breaking

Jun 29 – Jul 5 • 10 hrs

Capstone topic #1. Densest week – ~100 pp. Strategy: Mann 22 (setup) then attack Mann 23 (derivation). Tue pairs LB 46/47 (QFT voice) with Thomson 17 (LHC/pedagogy voice). **Mann 23 §23.2 is the single most important passage of the plan.**

MON

Mon Jun 29 • 4h • reading

Mann Ch. 22

Electroweak Unification

14 pp • ~1.5h

Mann Ch. 23

★ Higgs Mechanism (§§23.1–23.2)

~15 pp • ~2.5h

TUE

Tue Jun 30 • 4h • reading

Mann Ch. 23

★ Higgs Mechanism (§§23.3+)

~15 pp • ~1.5h

LB Ch. 46

Non-abelian Gauge Theory

9 pp • ~0.75h

LB Ch. 47

Weinberg-Salam Model

11 pp • ~1h

Thomson Ch. 17

Higgs (§§17.1, 17.3–17.5) [skim]

~18 pp • ~0.75h

WED

Wed Jul 1 • 2h • problems

Mann Ch. 22

Q2

Mann Ch. 23

★ Q1 and Q3

LB Ch. 46

Exercise (46.1)

LB Ch. 47

Exercise (47.1)

END-OF-WEEK CHECK – HIGGS

Derive the W and Z masses from the Higgs VEV ($v \approx 246$ GeV) and the weak mixing angle. Explain in plain English why the photon stays massless. If both work, you own the Higgs mechanism.

WK 10 ★ Neutrino Oscillations & Capstone

Jul 6 – Jul 11 • 10 hrs

Capstone topic #2. ~86 pp. Thomson 13 is the clearest pedagogical treatment; pair it with Mann 25 §§25.1–25.3 for the beyond-SM framing. Mann 24 ties up Wk 9’s electroweak machinery. **Thu Jul 9 – Sat Jul 11 are the capstone-write buffer days.**

MON

Mon Jul 6 • 4h • reading

Mann Ch. 24

Testing Electroweak Theory

16 pp • ~1.5h

Thomson Ch. 13

★ Neutrino Oscillations (§§13.1–13.5)

~22 pp • ~2.5h

TUE

Tue Jul 7 • 4h • reading

Thomson Ch. 13

★ Neutrino Osc. (§§13.6–13.9)

~12 pp • ~1.5h

Mann Ch. 25

★ Beyond SM (§§25.1–25.3)

~20 pp • ~2h

Thomson §17.8

Addendum: Neutrino Masses

3 pp • ~0.5h

WED

Wed Jul 8 • 2h • problems

Thomson Ch. 13

★ Problems 13.1, 13.3, 13.5

Mann Ch. 25

★ Q1

Mann Ch. 24

Q1

END-OF-WEEK CHECK – NEUTRINO OSC.

Derive the two-flavour vacuum oscillation probability $P(\nu_e \rightarrow \nu_\mu)$ from mixing angle θ and Δm^2 . Explain (i) why massless neutrinos can’t oscillate; (ii) why the solar-neutrino deficit led to the discovery of oscillations.

CAPSTONE BUFFER – THU JUL 9 / FRI JUL 10 / DEADLINE SAT JUL 11

Write the 1-page capstone connecting *gauge invariance* → *electroweak unification* → *Higgs mechanism* → *neutrino masses*. If you can draft it without referring back to the books, you own the material. Save final PDF and file to the sPHENIX/research notebook by end-of-day Saturday.

Progress Tracker (print & tick off)

One row per chapter. Capstone weeks tinted gold (Higgs) and green (neutrino oscillations); ★ rows are the two requested anchor chapters.

Wk	Chapter / item	Read	Probs	Notes
1	Loveland 8 – Beta Decay	☐	☐	
1	Loveland 9 – Gamma-Ray Decay	☐	☐	
2	Loveland 10 – Nuclear Reactions	☐	☐	
3	Loveland 11 – Fission	☐	☐	
3	Loveland 12 – Nuclear Astrophysics	☐	☐	
4	Loveland 14 – Reactors & Accelerators	☐	☐	
4	Loveland 18 – Radiation Detectors	☐	☐	
5	Mann 3 – Symmetries	☐	☐	
5	Mann 4 – Conservation Laws	☐	☐	
5	Mann 5 – Particle Classification	☐	☐	
5	Mann 6 – Discrete Symmetries	☐	☐	
5	Mann 9 – Scattering	☐	☐	
5	Mann 10 – A Toy Theory	☐	☐	
6	LB 7 – Example Lagrangians	☐	☐	
6	LB 12 – Canonical Quantization Ex.	☐	☐	
6	LB 13 – Many Components / Massive EM	☐	☐	
6	LB 16 – Propagators & Green's Fns.	☐	☐	
6	LB 17 – Propagators & Fields	☐	☐	
6	LB 18 – The S-matrix	☐	☐	
6	LB 19 – Feynman Diagrams	☐	☐	
6	LB 20 – Scattering Theory	☐	☐	
7	Mann 11 – Wave Equations	☐	☐	
7	Mann 12 – Gauge Invariance	☐	☐	
7	Mann 13 – QED	☐	☐	
7	Mann 14 – Testing QED	☐	☐	
8	Mann 19 – Beta Decay $\rightarrow$ Weak	☐	☐	
8	Mann 20 – Charged Leptonic Weak	☐	☐	
8	Mann 21 – Charged Weak of Quarks/Leptons	☐	☐	
8	LB 26 – Broken Symmetry	☐	☐	
9	Mann 22 – Electroweak Unification	☐	☐	
9	★ Mann 23 – Higgs Mechanism	☐	☐	
9	LB 46 – Non-abelian Gauge Theory	☐	☐	
9	LB 47 – Weinberg-Salam Model	☐	☐	
9	Thomson 17 – Higgs Boson (supp.)	☐	☐	
10	Mann 24 – Testing Electroweak	☐	☐	
10	★ Thomson 13 – Neutrino Oscillations	☐	☐	
10	★ Mann 25 – Beyond SM / ν -Osc.	☐	☐	
10	Thomson 17.8 – Neutrino Masses	☐	☐	
–	★ 1-page capstone (Sat Jul 11)	☐	—	

Slip Plan

If a Wed problem session runs out of time, prioritize: (1) ★ starred problems first, (2) problems from the chapter with the biggest page load, (3) everything else.

If a whole week slips, the safe drops are **Loveland 14**, **Mann 14**, **LB 26**, **LB 46**, and **Thomson 17 (supplementary)**. **Never drop** Mann 12, Mann 23, Mann 25, or Thomson 13 – those are the weeks' anchors.

Buffer rule: the lighter Loveland weeks (Wks 1–3) intentionally have spare hours – if Wks 5–8 (Mann/LB) start to slip, pull forward into the Loveland weeks rather than into Wk 9 (Higgs), which is already the densest. The Sat Jul 11 deadline is the hard backstop; the Thu–Sat days at the end of Wk 10 are reserved for the capstone write-up.